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APPLIED DATA SCIENCE &
ARTIFICIAL INTELLIGENCE



UNIVERSITÀ DEGLI STUDI DI TRIESTE

Ph.D. in Applied Data Science & Artificial Intelligence

XXXVIII cycle

Towards Flexible and Expressive Generative Models for Tabular and Relational Data

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Introduction

Publications

These are the publications (or submissions) that form the basis of this thesis:

1. **Zero-Shot Conditioning of Score-Based Diffusion Models by Neuro-Symbolic Constraints.** Scassola, D., S. Saccani, G. Carbone, and L. Bortolussi. [Accepted at AAAI 2025](#).
2. **Graph-Conditional Flow Matching for Relational Data Generation.** Scassola, Davide, Sebastiano Saccani, and Luca Bortolussi. [Accepted at AAAI 2026](#).
3. **A Sobering Look at Tabular Data Generation with Probabilistic Circuits.** Scassola, D., Ponsford, D., et al. [Submitted to UAI 2026](#).

Other publications:

- *"Model Abstraction and Conditional Sampling with Score-Based Diffusion Models."* Bortolussi, Luca, et al. International Conference on Quantitative Evaluation of Systems (2023).
- *"Multi-objective reward generalization: improving performance of Deep Reinforcement Learning for applications in single-asset trading."* Cornalba, Federico, et al. Neural Computing and Applications (2024).

Generative Models for Tabular Data

- Generative modeling has advanced rapidly, its development mostly focused on unstructured data such as images, videos, audio, and text.
- **Tabular data generation** is less popular despite its practical importance (financial transactions, patient records, commercial databases, etc.)
- Tabular data is structured data organized in rows and columns, where each row is an observation and each column is a feature or attribute.

age	education	sex	capital-gain	native-country	income
39	Bachelors	Male	2174	United-States	<=50K
50	Bachelors	Male	0	Jamaica	<=50K
37	Masters	Female	14084	United-States	>50K
...	...	...	...	...	...

- **Challenges:** complex marginals, intricate dependencies, heterogeneous data types, missing data, hard constraints, quality evaluation, privacy concerns, relational data.

Applications of Tabular Data Generation

- **Privacy preservation:** share synthetic data that mimics the statistical properties of real data without revealing sensitive information.
- **Data augmentation:** create additional training data to improve the performance of machine learning models, especially in cases of class imbalance or limited data availability.
- **Probabilistic queries:** perform probabilistic reasoning on the generated data, e.g., computing probabilities (anomaly detection).



Limitations of Current Techniques

Despite significant progress, recently driven by powerful diffusion models, several limitations remain, especially regarding flexibility:

- **Constraints:** It is hard to efficiently incorporate user-specified constraints in the generated data (known facts, partial observations, logical rules).
- **Relational data:** It is hard to generate realistic relational datasets (multiple tables with foreign keys).
- **Probabilistic queries:** General probabilistic queries are generally not possible with most state-of-the-art models for tabular data generation: density computation, marginalization, inference under missing data, etc.

This thesis aims at answering the following questions:

- **Q1:** Can we generate data from a pre-trained diffusion model while enforcing arbitrary user-defined logical constraints?
- **Q2:** Can we design expressive generative models capable of generating relational data with complex structural dependencies?
- **Q3:** Can we train tractable yet expressive generative models for tabular data?

Contributions:

1. Method for conditional sampling from score-based diffusion models under logical constraints
2. Flow-matching framework for relational databases with arbitrary structures
3. Analysis of probabilistic circuits as tractable generative models

Training-free Conditioning of Score-Based Generative Models by Neuro-Symbolic Constraints

Controlling Generative Models

A relevant problem in generative models is to efficiently achieve conditional generation by forcing user-defined logical constraints on generated samples.

- Example: *"Given a generative model for tabular data with 3 fields $p(x_1, x_2, x_3)$, sample from $p(x_1, x_2, x_3 \mid x_1 > x_2 \wedge x_3 = 2)$ "*
- Example: *"Given a generative model for MNIST images of digits, generate digits that satisfy horizontal symmetry"*

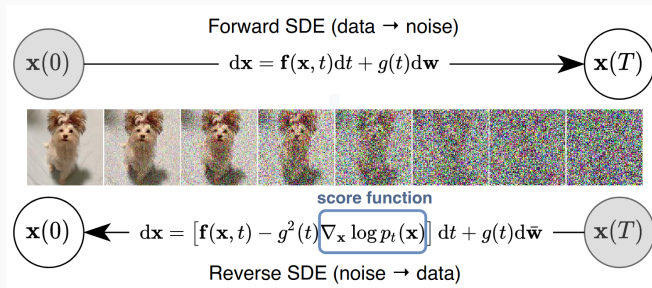
Directly learning a conditional model $p(\mathbf{x} \mid \mathbf{y})$ for any condition $\mathbf{y}$ is not feasible.

Q1: Can we generate data from a pre-trained diffusion model while enforcing arbitrary user-defined logical constraints?

Background: Score-Based Diffusion Models

Objective: generate new samples from the distribution $p(\mathbf{x})$ that generated the data

1. Define a diffusion process $p(\mathbf{x}_t|\mathbf{x}_0)$ that gradually adds noise to clean data points $\mathbf{x}_0$ until pure noise is reached (as a standard Gaussian).
2. Learn the noise-dependent score: $\mathbf{s}(\mathbf{x}_t, t) := \nabla_{\mathbf{x}_t} \ln p(\mathbf{x}_t)$, for example, by **denoising score matching**.
3. Run a sampling technique that uses the score to sample new points by reversing the diffusion process.



- **Problem:** Given the unconditional score $\nabla_{\mathbf{x}_t} \ln p(\mathbf{x}_t)$ and a logical constraint $\pi(\mathbf{x}) : \mathbb{R}^d \rightarrow \{0, 1\}$, sample from

$$p(\mathbf{x}|\pi) \propto p(\mathbf{x})\pi(\mathbf{x})$$

- **Continuous relaxation:** Given a differentiable constraint $c(\mathbf{x}) : \mathbb{R}^d \rightarrow (-\infty, 0]$ expressing the degree of satisfaction, sample from

$$p_c(\mathbf{x}) \propto p(\mathbf{x})e^{c(\mathbf{x})}$$

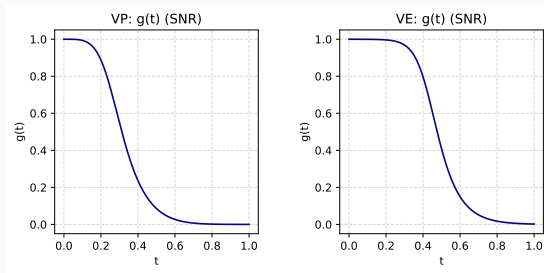
Conditional Score Approximation

We approximate the conditional score by interpolating the two regimes where the conditional score is known without approximation (no noise $t = 0$, only noise $t = 1$):

$$\tilde{s}_c(\mathbf{x}_t, t) = \nabla_{\mathbf{x}_t} \ln p(\mathbf{x}_t) + g(t) \nabla_{\mathbf{x}_t} c(\mathbf{x}_t)$$

where $g(0) = 1$ and $g(1) = 0$.

We use as weighting function $g(t)$ the signal-to-noise ratio of the diffusion kernel.



Intuition: the gradient of the constraint is more meaningful as the data gets less noisy.

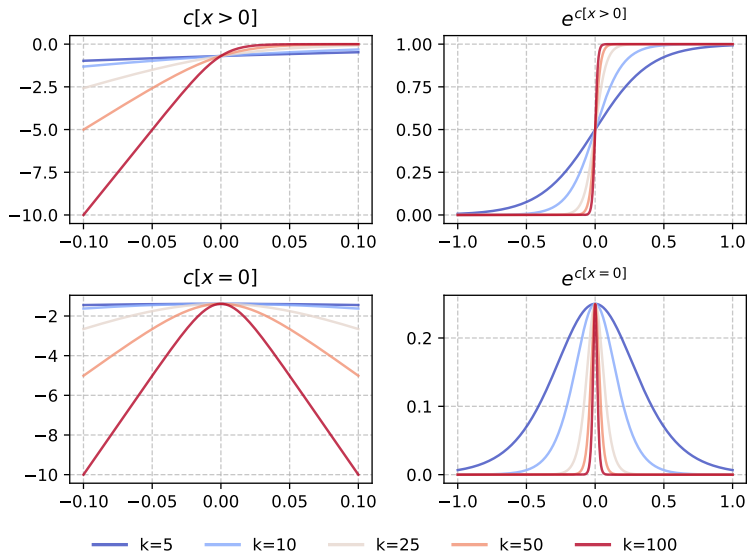
Neuro-Symbolic Logical Constraints

- The score approximation needs a differentiable constraint $\rightarrow$ we relax logical constraint into soft, differentiable constraints.
- Every atomic predicate and logical connector is associated to a differentiable function (similar to Badreddine et al. [2023]):

Atom	Function	Connector	Function
$x \geq y$	$-\ln(1 + e^{k(y-x)})$	$a \wedge b$	$a + b$
$x = y$	$x \geq y \wedge x \leq y$	$a \vee b$	$\ln(e^a + e^b - e^{a+b})$
		$\neg a$	$\ln(1 - e^a)$

- Connectors are based on probabilistic operations in log-space.
- This set of functions has desirable properties: exact NNF (De Morgan's laws), numerical stability, tunable hardness (k).

Neuro-Symbolic Logical Constraints



- **Data:**

- Tabular data: White wine (11 features) dataset and Adult (15 features)
- Time series: simulated eSIRS time series (spreading of a disease in an open population)
- Images: MNIST and CelebA

- **Baseline:**

- Rejection Sampling (RS) as ground truth (when possible)
- Universal Guidance [Bansal et al., 2023]

- **Metrics:**

- Total variation distance: $D(X, Y) := \frac{1}{2} \sum_i |x_i - y_i|$ where x_i and y_i are the empirical probabilities for a given common binning of samples X and Y .
- Average l1 distance in correlation coefficients.
- Satisfaction of the constraint (% of samples).

Experiments: Tabular Data

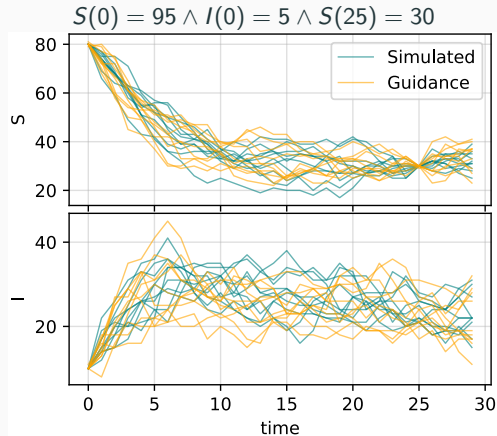
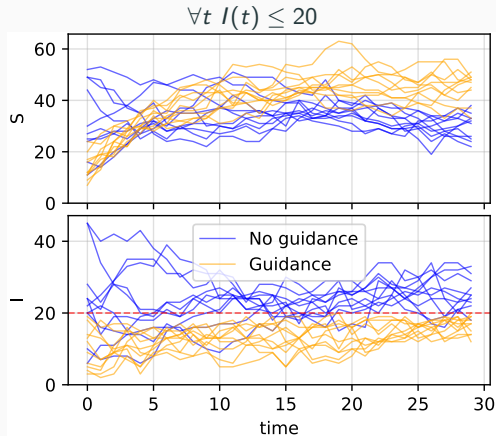
1. **White wine dataset:** $(\text{fixed acidity} \in [5.0, 6.0] \vee \text{fixed acidity} \in [8.0, 9.0]) \wedge \text{alcohol} \geq 11.0 \wedge (\text{residual sugar} \leq 5.0 \rightarrow \text{citric acid} \geq 0.5)$
2. **White wine dataset:** $\text{alcohol}_1 > \text{alcohol}_2 + 1$
3. **Adult dataset:** $\text{age} \geq 40 \wedge (\text{race} \neq \text{"White"} \vee \text{education} = \text{"Masters"})$

Experiment	Avg TV	Avg L1 corr	Hard sat	RS acceptance rate
White wine	0.1	0.07	86%	1.67%
White wine (multi)	0.061	0.027	99%	27%
Adult	0.067	0.046	100%	1.73%

- Low error for simple constraints, still acceptable for complex constraints.
- More efficient sampling in low probability regions.

Experiments: Time Series

Dataset: Simulated eSIRS time series (spreading of a disease in an open population). Fixed population of size N composed of Susceptible (S), Infected (I), and Recovered (R) individuals.



- High satisfaction rate, low error (When RS was possible)

Experiments: Images

Image symmetry:

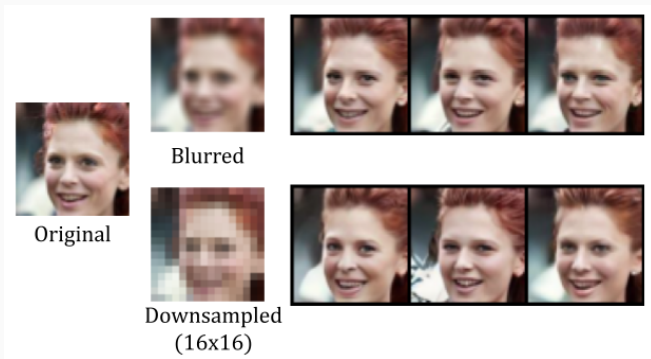


We had to trade-off sample quality and constraint satisfaction by tuning the constraint strength k .

Experiments: Images

Image restoration (with known corruption process f):

$$\forall i \ f(\mathbf{x})_i = \tilde{\mathbf{y}}_i$$



Comparison with Universal Guidance [Bansal et al., 2023]

- Synthesis of previous optimization-based approaches, originally applied to images
- It fails in two of the tasks we tested (White wine and eSIRS bridging).

Method	White wine		eSIRS bridging	
	Avg TVD	Max TVD	Avg TVD	Max TVD
Ours	0.1	0.15	0.13	0.25
Univ. Guidance	0.29	0.85	0.47	1.0

Contributions

- We developed a method for training-free conditional distribution approximation with logical constraints, that can be applied to any pre-trained score-based diffusion model.
- Our method performs well on training-free conditional distribution approximation on tabular data with logical constraints.

Limitations

- As complexity increases, it is necessary to trade off constraint satisfaction with sample quality.
- Optimization-based methods are preferable when it is more important to obtain high-quality samples, rather than approximating the true conditional distribution.

Graph-Conditional Flow Matching for Relational Data Generation

The Problem

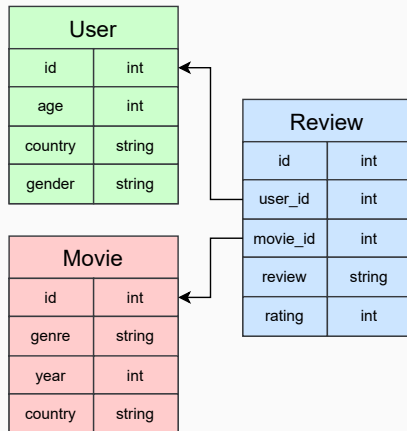
Q2: Can we design expressive generative models capable of generating relational data with complex structural dependencies?

Generating synthetic relational data:

- Multiple tables (structured, relatively low-dimensional, heterogeneous data)
- Foreign-key relationships between records

Why?

- Privacy: share useful synthetic data while preserving privacy
- Data augmentation

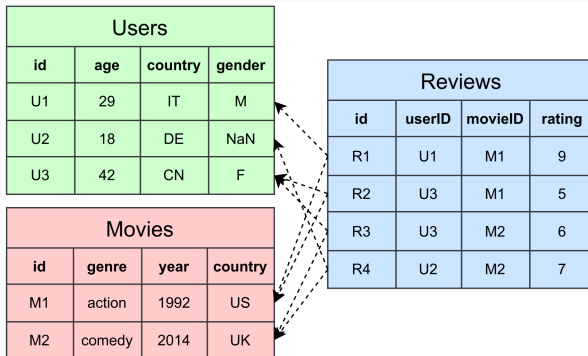


Generating Relational Data is Hard

Generating relational data requires modeling statistical dependencies between records connected by foreign keys (also indirectly).

Most existing methods show:

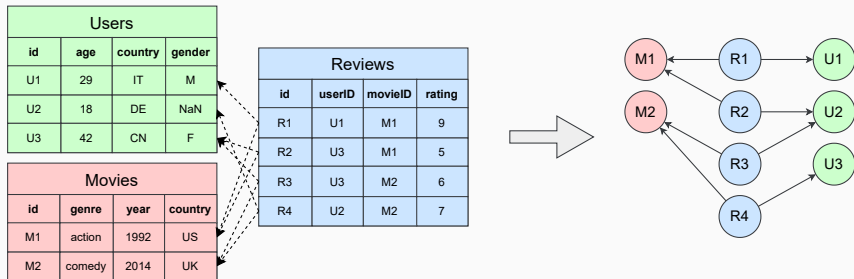
- Limited expressiveness (independence assumptions, heuristics)
- Limited flexibility with arbitrary foreign-key structures



Relational Data is a Graph

A relational database can be represented as a large graph:

- Records $\rightarrow$ nodes
- Foreign-key relationships $\rightarrow$ edges



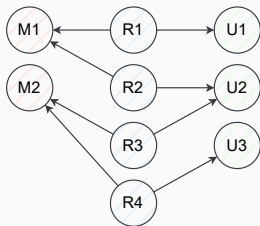
- The foreign-key graph can be large and complex.
- The i.i.d. samples are the connected components of the graph.

Graph-Conditional Generation with Flow Matching

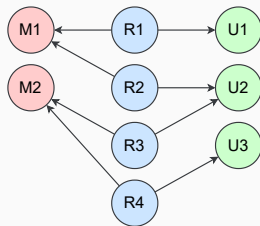
Our approach: generate whole connected components conditioning on the foreign-key graph.

1. Generate the foreign-key graph $G \sim p(G)$ using a scalable method
2. Use flow matching (diffusion-like generative model) for generating whole connected components (X, G) by “filling” the empty foreign-key graph G with content X

$$p(G, X) = p(G)p(X | G)$$



$G \sim p(G)$
(scalable graph model)

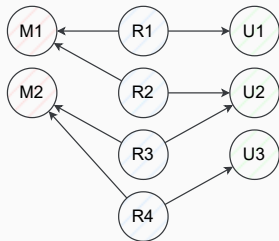


$X \sim p(X | G)$
(deep generative model)

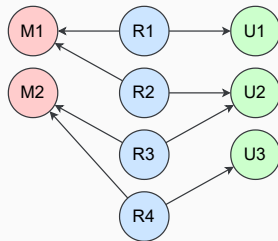
Graph-Conditional Flow Matching

This approach has advantages:

- **Expressive**: records in a connected component are modeled jointly
- **Flexible**: any foreign-key graph can be used
- **Scalable**: flow matching scales to large dimensionalities + no need for materializing the dense adjacency matrix



$G \sim p(G)$
(scalable graph model)



$X \sim p(X | G)$
(deep generative model)

(Variational) Flow Matching [Lipman et al., 2023, Eijkelboom et al., 2024]

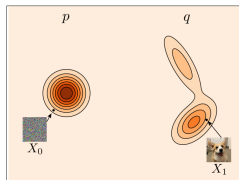
Flow Matching = Continuous Normalizing Flows (ODEs) + diffusion-like training

- **Sampling:** solve the ODE to map Gaussian noise to data:

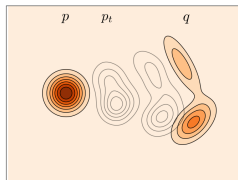
$$\frac{d}{dt} \varphi_t(\mathbf{x}) = v_t(\varphi_t(\mathbf{x})), \quad \text{with initial condition } \varphi_0(\mathbf{x}) = \mathbf{x}.$$

- **Training:** learn field v_t by **denoising** samples from a chosen **conditional probability path**:

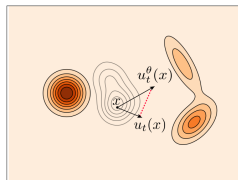
$$\mathcal{L}(\theta) = -\mathbb{E}_{t \sim \mathcal{U}(0,1), \mathbf{x}_1 \sim p_{\text{data}}, \mathbf{x}_t \sim p_t(\mathbf{x}_t | \mathbf{x}_1)} [\log p_{\theta}(\mathbf{x}_1 | \mathbf{x}_t)] \quad v_t^{\theta}(\mathbf{x}_t) = \mathbb{E}_{\mathbf{x}_1 \sim p_{\theta}(\mathbf{x}_1 | \mathbf{x}_t)} [u_t(\mathbf{x}_t | \mathbf{x}_1)]$$



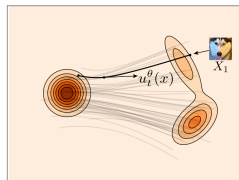
(a) Data.



(b) Path design.



(c) Training.

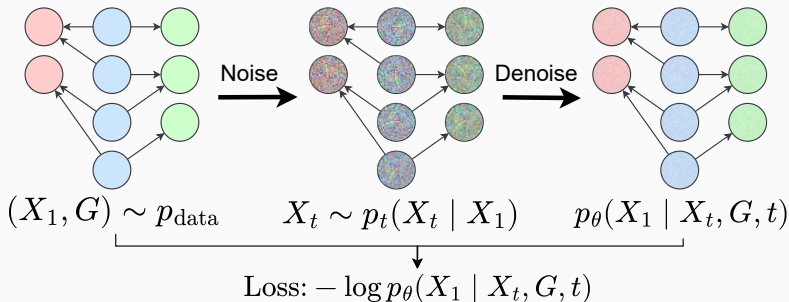


(d) Sampling.

Flow Matching for Relational Data: Training

Training:

- Define a conditional probability path $p_t(X_t | X_1, G) = \prod p_t^i(x_t^i | x_1^i)$ independently for each feature.
- Learn a denoiser $p_\theta(X_1 | X_t, G) = \prod p_\theta^i(x_1^i | X_t, G)$
- Categoricals are one-hot encoded, and the denoiser distribution is either categorical or Gaussian depending on the feature type

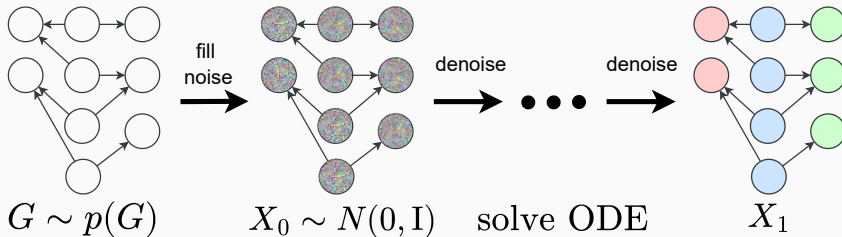


Flow Matching for Relational Data: Sampling

Sampling:

1. Generate the foreign-key graph $G \sim p(G)$ (we just re-sample connected components from the original graph)
2. Initialize every node with Gaussian noise $X_0 \sim \mathcal{N}(0, I)$
3. Solve the ODE numerically using the learned velocity, parameterized by the learned denoiser:

$$v_t^\theta(X_t) = \mathbb{E}_{\mathbf{x}_1 \sim p_\theta(\mathbf{x}_1 | X_t, G)} [u_t(X_t | \mathbf{x}_1)]$$



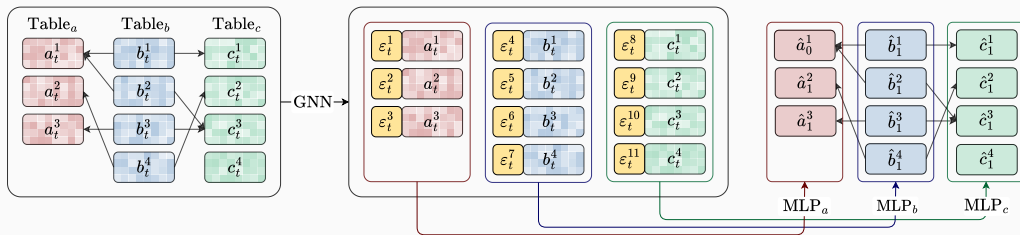
Denoiser Architecture

- A GNN computes node embeddings for each record

$$\varepsilon_t^i = \text{GNN}_\theta(X_t, G, t)^i$$

- MLPs use noisy records and node embeddings to predict the clean record

$$\hat{x}_1^i = \mathbb{E}_{p_\theta(x_1^i | X_t, G)} [x_1^i] = \text{MLP}_\theta(x_t^i, \varepsilon_t^i, t)$$



Results

- We use the *SyntheRela* benchmark library [Hudovernik et al., 2024] for metrics and benchmarks.
- Fidelity is measured as the accuracy of an XGBoost classifier, trained to distinguish real from generated records (the lower the better).

	AirBnB	Biodegradability	CORA	IMDB	Rossmann	Walmart
Ours	0.58 ± 0.03	0.59 ± 0.02	0.63 ± 0.02	0.59 ± 0.03	0.51 ± 0.01	0.73 ± 0.01
Ours (no GNN)	0.70 ± 0.005	0.86 ± 0.004	0.62 ± 0.004	0.89 ± 0.002	0.75 ± 0.01	0.91 ± 0.04
Hudovernik [2024]	0.67 ± 0.003	0.83 ± 0.01	0.60 ± 0.01	0.64 ± 0.01	0.77 ± 0.01	0.79 ± 0.04
ClavaDDPM	≈ 1	-	-	0.83 ± 0.004	0.86 ± 0.01	0.74 ± 0.05
RCTGAN	0.98 ± 0.001	0.88 ± 0.01	0.73 ± 0.01	0.95 ± 0.002	0.88 ± 0.01	0.96 ± 0.02
REaLTabF.	≈ 1	-	-	-	0.92 ± 0.01	≈ 1
SDV	≈ 1	0.98 ± 0.01	≈ 1	-	0.98 ± 0.003	0.90 ± 0.03

- We achieve state-of-the-art performance
- **Privacy:** DCR (Distance to Closest Record) analysis does not highlight privacy leaks.
Intuitively, we check if generated records are too similar to real records in the training set.

Our Contribution

Expressiveness

- We model whole connected components (no independence assumptions) with flow matching + GNNs for handling graphs.
- We achieve generalization by controlling “within-sample” generalization and apply a bottleneck on the GNN embedding size.

Flexibility

- The GNN-based denoiser supports any kind of graph.

Scalability

- Flow matching scales with large dimensionalities.
- We avoid dealing with dense adjacency matrices.

Limitations

- Not a model of the foreign-key graph $p(G)$
- Hyperparameter tuning for each dataset

Future Work

- Better engineer the denoiser, e.g., more advanced GNN architectures
- Batching strategies to scale to larger connected components

Tabular Data Generation with Probabilistic Circuits

Q3: Can we train tractable yet expressive generative models for tabular data?

The Context:

- The landscape of tabular data generation is currently dominated by deep generative models.
- Unconditional sampling is often the only exact probabilistic query they can compute.
- The state of the art is represented by diffusion models, which often show "saturated" performance on standard benchmarks.

Idea: Use probabilistic circuits, which are tractable models, as tabular data generators.

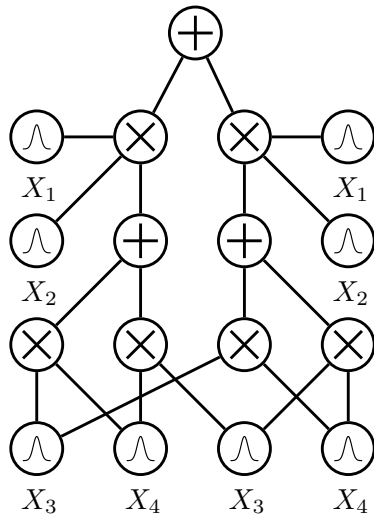
Probabilistic Circuits (Hierarchical Mixture Models)

Probabilistic circuits (PC) are tractable models where the density function $p(\mathbf{x})$ is defined as a computational graph composed of three types of nodes:

- **Input nodes:** tractable distributions $q_\theta(x)$ (ex: $\mathcal{N}(x \mid \mu, \sigma^2)$)
- **Sum nodes:** mixture of distributions (defined over the same variables) $p(\mathbf{x}) = \sum_i w_i q_i(\mathbf{x})$
- **Product nodes:** product of distributions (defined over different variables) $p(\mathbf{x}) = \prod_i q_i(x_i)$

Model parameters w_i and θ are learnable by maximum likelihood and gradient-based optimization.

The structure is a hyperparameter, usually built with heuristics.



Probabilistic circuits allow the following inference queries to be computed in time linear in the size of the circuit:

- **Density Evaluation** by bottom-up evaluation of the circuit
- **Sampling** by latent ancestral sampling
- **Marginalization** of any subset of variables. This is done by setting input units of marginalized variables to 1.
- **Conditional Sampling** conditioned on any subset of variables

Overparameterization

In order to build generative models for tabular data, we need to increase the expressiveness of probabilistic circuits by increasing the number of parameters.

- **Vectorized input layers:** For each variable use more input units of the same family of distributions.

$$\text{Ex: } \mathbf{p}(x) = \{\mathcal{N}(x \mid \mu_1, \sigma_1^2), \dots, \mathcal{N}(x \mid \mu_k, \sigma_k^2)\} \quad (1)$$

- **Sum-product layers:** Vectorized distributions of disjoint sets of variables are combined with sum-product layers, in particular we use the CP layer [Loconte et al., 2025a]:

$$\mathbf{p}(\mathbf{x}) \otimes \mathbf{q}(\mathbf{y}) = \left(Q_1 \mathbf{p}(\mathbf{x}) \right) \odot \left(Q_2 \mathbf{q}(\mathbf{y}) \right) \quad (2)$$

Sum unit $\rightarrow$ Matrix multiplication

Product unit $\rightarrow$ Hadamard product

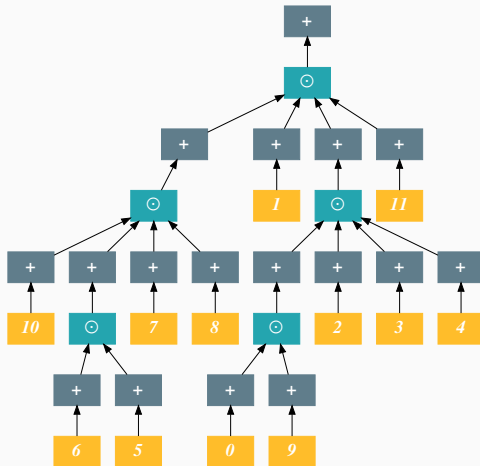
We build probabilistic circuits for tabular data (**TabPC**) following these design choices:

- Gaussian input layers for numeric variables, categorical input layers for categorical variables.
- CP-layer sum-product units
- Structure learned with the Chow-Liu algorithm. It mixes together highly dependent variables earlier in the circuit.
- Preprocessing for tabular data generation (quantile normalization, handling inflated values...).

The only hyperparameter is the "size" of the circuit.

Example

Example of TabPC built for the Magic dataset (12 features/columns).

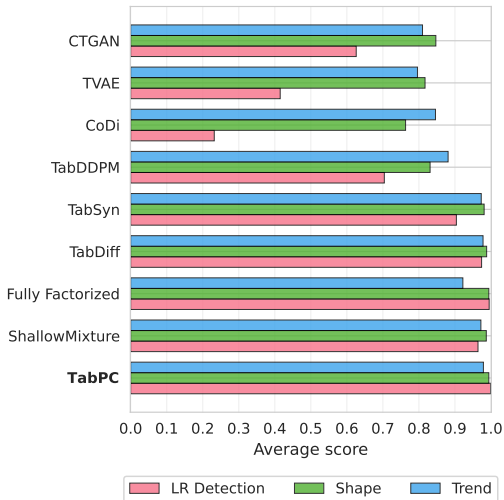


Experiments

We aim at comparing with state-of-the-art diffusion-based methods for tabular data generation. We run the experiments in the same settings as TabDiff [Shi et al., 2025]:

- **Datasets:** 6 common datasets from the UCI machine learning repository: Adult, Beijing, Default, Diabetes, Magic, News and Shoppers.
- **Baselines:**
 - Open-source models based on VAEs, GANs, LLMs, and diffusion.
 - Fully Factorized model (FF)
 - Shallow Mixture (SM)
- **Metrics:**
 - *Fidelity* metrics: Shape (quality of marginals), Trend (quality of correlations), Detection based on logistic regression
 - *Utility* metric: Machine Learning Efficiency (MLE);
 - *Privacy* metric: Distance to Closest Record (DCR).

Results: Shape, Trend and LR-Detection



We evaluate Shape, Trend and LR-Detection metrics and report the average over the 6 datasets. All these metrics have values between 0 and 1, where 1 is the best possible score.

- Several models saturate these metrics.
- With proper preprocessing, even a fully factorized model can achieve good performance on these metrics.

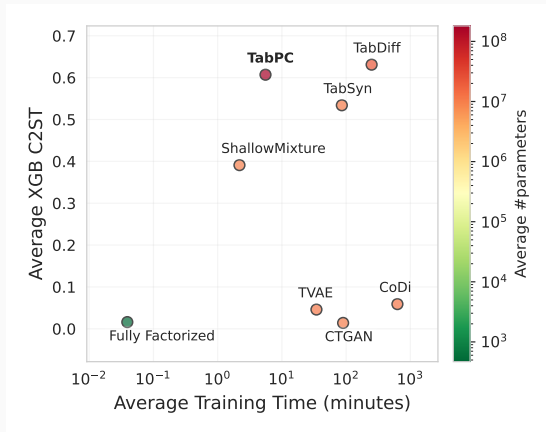
LR-Detection

- Logistic Regression is too easy to fool.
- We prove that a synthetic dataset able to replicate the mean of each feature is enough to achieve a perfect score.
- We suggest using a more powerful classifier or an ensemble of classifiers to make the detection more robust, as done in other works. We propose using XGBoost, which is powerful and fast to evaluate.

Trend

- Trend should measure the similarity of correlations, but it is heavily influenced by the quality of marginals.
- We propose wNMIE (weighted Normalized Mutual Information Error) which is independent of the quality of the marginals.

TabPC: Quality and Efficiency



- **Quality:** TabPC is competitive with state-of-the-art diffusion-based models.
- **Training time:** TabPC is significantly faster to train: $\approx 40 - 50\times$ faster than TabDiff.
- **Sampling time:** TabPC is also faster to sample from, $\approx 10\times$ faster than TabDiff.
- **Number of parameters:** TabPC requires more parameters, $\approx 10\times$ more than TabDiff.

- TabPC is competitive with state-of-the-art tabular data generation models from the point of view of quality of generated data (fidelity and utility).
- TabPC is faster to train and sample from.
- TabPC is tractable, allowing the **exact** computation of probabilistic queries that are not possible with other models:
 - Density evaluation
 - Marginalization (useful for handling missing data)
 - Conditional sampling

Limitations:

- TabPC struggles when the number of features is large and there are high correlations between them.
- TabPC requires more parameters.

Future work:

- Multivariate Gaussians as input units
- Variants of PCs: Probabilistic Integral Circuits [Gala et al., 2024], Sum-of-Squares Circuits [Loconte et al., 2025b], ...
- Expectation Maximization

Conclusion

Summary: Thesis Goals

- Recent progress: diffusion-like models for tabular data yield high fidelity and empirical evidence of privacy protection.
- Key limitations: limited conditional flexibility, intractable probabilistic queries, and immature relational data generation.
- Thesis goal: address these limitations with tractable, flexible, and relationally-aware generative modeling.

Summary: Methods and Findings

1. **Training-free conditional sampling with logical constraints for score-based models** → strong results but challenges with complex/high-dimensional constraints.
2. **Graph-conditional flow matching for relational data generation** → improved expressiveness and flexibility for relational data generation.
3. **Overparametrized probabilistic circuits for tabular data** → competitive fidelity, exact probabilistic queries and faster training/sampling.

Open challenges:

- Increasing the power of training-free conditioning methods
- Generating realistic relational datasets with foreign-key graphs
- Improving the expressiveness of probabilistic circuits for tabular data

Thank you! Questions?

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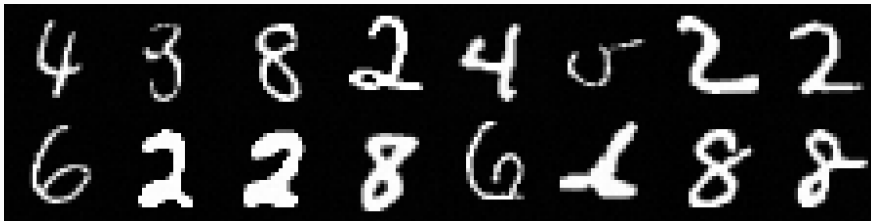
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Experiments: Images

MNIST digits sum: *"Sample two MNIST images whose digit labels sum up to 10"*

$$\bigvee_{i=1}^9 \text{class}(\mathbf{x}, i) \wedge \text{class}(\mathbf{y}, 10 - i)$$

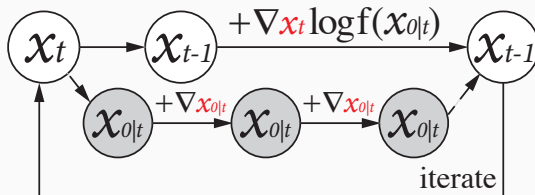
Where "class" is a pretrained classifier.



According to the classifier $\approx 96\%$ of pairs sum up to 10, but only 2-8, 6-4 pairs are generated.

Comparison with Universal Guidance

- Universal Guidance (Bansal et al. [2023]) is probably the state of the art for plug-and-play conditioning.
- Originally applied to imaging problems
- Synthesis of previous optimization-based approaches for solving inverse problems with diffusion models
- Oversimplifying, it involves at each time step an optimization of the denoising prediction $\mathbf{x}_0$ (that is a linear transformation of the score), in order to make it satisfy the constraint $c(\mathbf{x}_0)$.



To sample from the target $p_c(\mathbf{x}) \propto p(\mathbf{x})e^{c(\mathbf{x})}$, we need the conditional noise-dependent score $\nabla_{\mathbf{x}_t} \ln p_c(\mathbf{x}_t)$:

- At $t = 0$ (no noise $\rightarrow$ analytical form)

$$\nabla_{\mathbf{x}_0} \ln p_c(\mathbf{x}_0) = \nabla_{\mathbf{x}_0} \ln p(\mathbf{x}_0) + \nabla_{\mathbf{x}_0} c(\mathbf{x}_0)$$

- At $t = 1$ (only noise $\rightarrow$ analytical form)

$$\nabla_{\mathbf{x}_1} \ln p_c(\mathbf{x}_1) = \nabla_{\mathbf{x}_1} \ln p(\mathbf{x}_1)$$

- When $t \in (0, 1)$ there is no general analytical form $\rightarrow$ approximation

Comparison with Universal Guidance

Universal guidance fails in two of the tasks we tested (White wine and eSIRS bridging). However, it performs better on one of the tasks based on images.

Method	White wine		eSIRS bridging	
	Avg TV	Max TV	Avg TV	Max TV
Ours	0.1	0.15	0.13	0.25
Univ. Guidance	0.29	0.85	0.47	1.0

Comparison with Universal Guidance

Universal Guidance performs better on image problems:

$$\bigvee_{i=1}^9 \text{class}(\mathbf{x}, i) \wedge \text{class}(\mathbf{y}, 10 - i)$$



It is able to generate high quality samples but it misses the correct approximation of the conditional distribution, probably because it involves an optimization phase.

Training-Free Conditional Sampling

Input: pre-trained score-based diffusion model $\nabla_{\mathbf{x}_t} \ln p(\mathbf{x}_t)$ for $p(\mathbf{x})$ and logical constraint π .

1. Convert the logical constraint to a differentiable constraint using the neuro-symbolic framework:

$$\pi(\mathbf{x}) \rightarrow c(\mathbf{x}) \quad (3)$$

2. Approximate the conditional score

$$\tilde{\mathbf{s}}_c(\mathbf{x}_t, t) = \nabla_{\mathbf{x}_t} \ln p(\mathbf{x}_t) + g(t) \nabla_{\mathbf{x}_t} c(\mathbf{x}_t) \quad (4)$$

3. Substitute the new score function in the original score-based sampling algorithm to generate new points

No training required!

Datasets Details (1)

Table 1: Overview of the datasets used in the experiments, showing for each table of each dataset the number of rows, the number of features (feature columns) and tables referred by foreign keys (the parent tables).

Dataset	Table	# Rows	# Features	Foreign Keys
AirBnB	users	10,000	15	–
	sessions	47,217	5	users
Biodegradability	molecule	328	3	–
	group	1,736	1	–
	atom	6,568	1	molecule
	gmember	6,647	–	atom, group
	bond	6,616	1	atom1, atom2
CORA	paper	2,708	1	–
	content	49,216	1	paper
	cites	5,429	–	paper1, paper2

Datasets Details (2)

Table 2: Overview of the datasets used in the experiments, showing for each table of each dataset the number of rows, the number of features (feature columns) and tables referred by foreign keys (the parent tables).

Dataset	Table	# Rows	# Features	Foreign Keys
IMDB MovieLens	users	6,039	3	–
	movies	3,832	4	–
	actors	98,690	2	–
	directors	2,201	2	–
	ratings	996,159	1	movie, user
	movies2actors	138,349	1	movie, actor
	movies2directors	4,141	1	movie, director
Rossmann	store	1,115	9	–
	historical	57,970	7	store
Walmart	stores	45	2	–
	features	225	11	store
	depts	15,047	4	store

Core Approach:

- GNNs compute node embeddings ϵ^i for each record i
- Handles heterogeneous graphs with multiple node/edge types
- Edge-type-specific convolutions aggregate neighbors by table type

Two Architecture Variants:

GATv2-based: Brody et al. [2021]

- Attention mechanism
- Residual connections
- Hidden dim: 100
- Used for: AirBnB, Rossmann, Walmart

GIN-based: Xu et al. [2018]

- 3 GIN layers
- MLP width: 100
- Embedding size: 20 or 50
- Used for: CORA, IMDB, Biodegradability

Architecture:

- Each table k has a dedicated MLP denoiser f^k that predicts clean records:

$$\hat{\mathbf{x}}_1^i = f^{k_i}(\mathbf{x}_t^i, t, \epsilon_t^i)$$

- **Inputs:** noisy record $\mathbf{x}_t^i$ + time embedding t + node embedding ϵ_t^i
- **Design:** Layer Normalization + SiLU activations + linear output
- **Output:** Softmax for categoricals, linear for continuous features

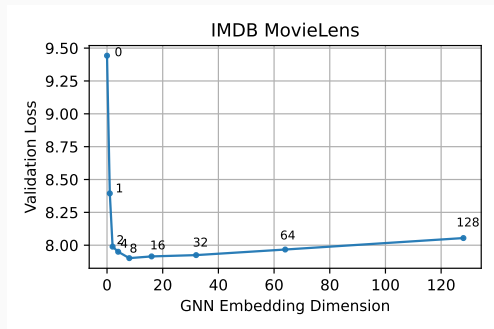
Hyperparameters:

- 2-3 hidden layers per table
- Width: 10-1000 hidden units (dataset-specific)

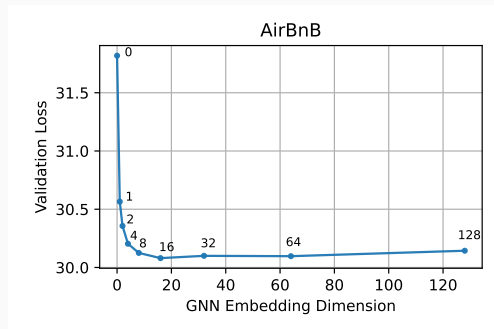
Tuning Node Embedding Size

Key Tradeoff:

- Too large → memorization and privacy leaks
- Too small → limited expressiveness
- Our choice: 2-10 dimensions (conservative for privacy)



(a) IMDB MovieLens



(b) Airbnb

Results: Privacy and Efficiency

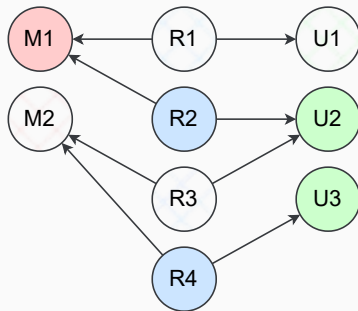
- **Privacy:** DCR (Distance to Closest Record) analysis does not highlight privacy leaks. Intuitively, we check if generated records are too similar to real records in the training set.
- **Efficiency:** Training and generation are fast. ¹

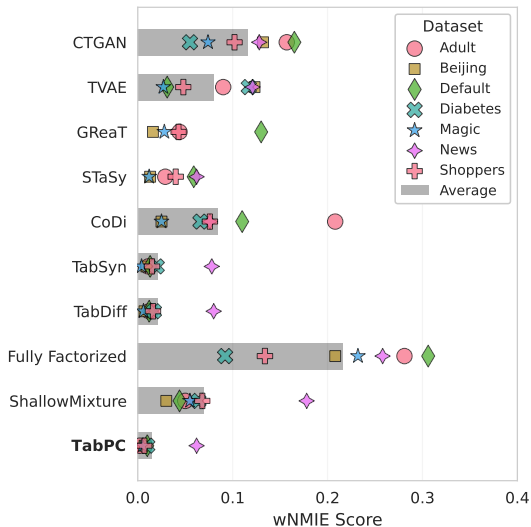
Dataset Name	Running Time
AirBnB	10m 3s
Biodegradability	1m 6s
CORA	3m 10s
IMDB MovieLens	14m 25s
Rossmann	2m 57s
Walmart	1m 48s

¹on a single NVIDIA RTX A5000

Controlling Generalization Error

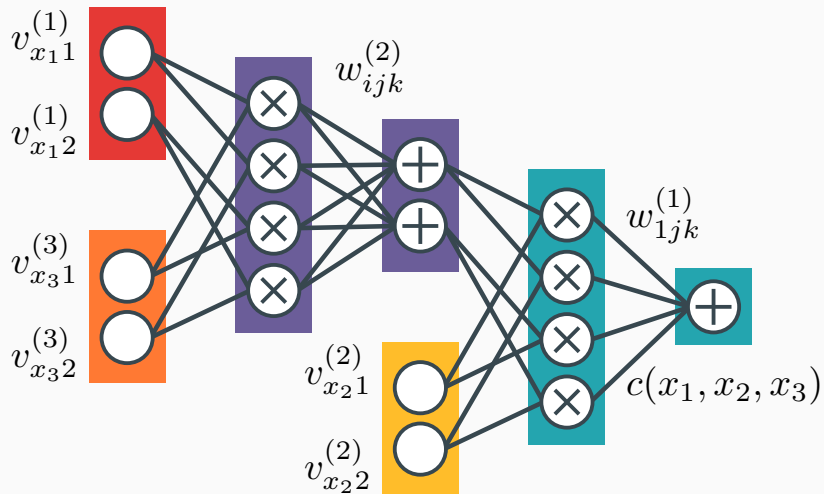
- We randomly split the nodes between train and validation nodes.
- “Within sample” generalization $\rightarrow$ no need for training with many i.i.d. connected components!
- Bottleneck on the GNN embedding size $\rightarrow$ prevents graph memorization





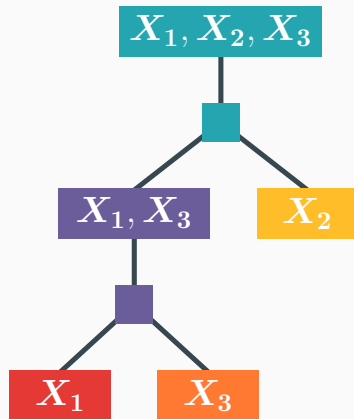
- wNMIE is not influenced by marginals, fully factorized models perform poorly as expected.
- TabPC is competitive with state-of-the-art diffusion-based models.

Example of Overparameterized Circuit



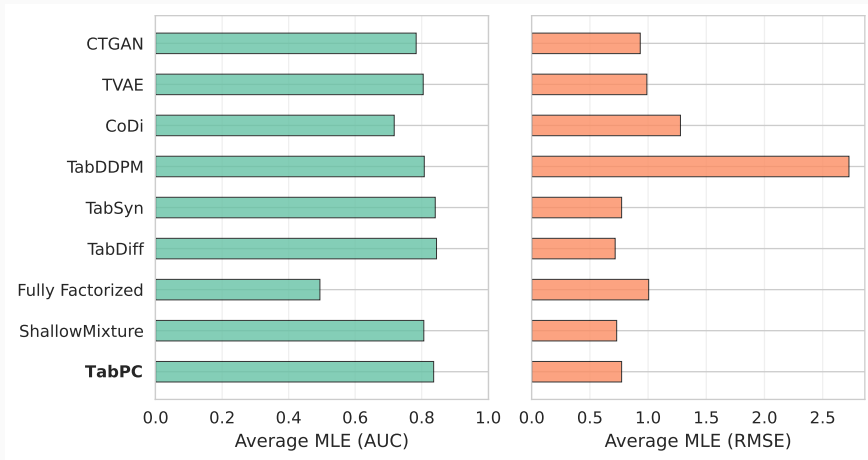
Structure Learning

- The graph defining the order sets of variables are mixed together is called the region graph.
- We use a method based on the Chow-Liu algorithm [Vergari et al., 2015, Rahman et al., 2014, Choi et al., 2011] to learn the structure of the region graph from data.
- The Chow-Liu algorithm is an efficient method for constructing a tree-shaped Bayesian network approximating a joint distribution, from which a dataset of samples is available.
- In such a region graph, highly dependent variables are mixed together earlier.

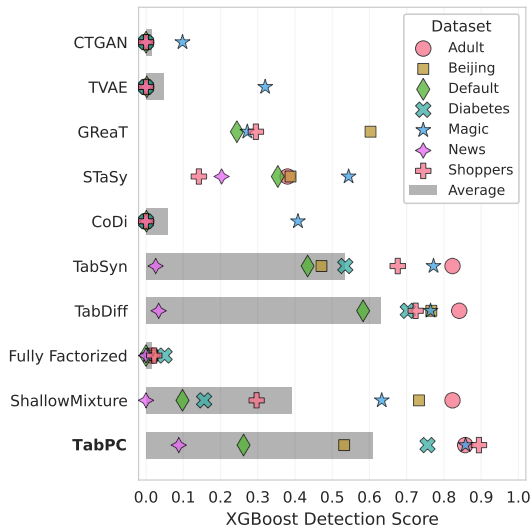


Results: MLE

Machine Learning Efficiency (MLE) is low for the fully factorized model, but there is also some saturation effect.



XGBoost-Detection



- XGBoost better distinguishes between models.
- TabPC is competitive with state-of-the-art diffusion-based models.
- TabPC still struggles for some datasets, probably due to the large number of features and high correlations.

Table 3: DCR scores (%). In this table we report the percentage of times the DCR of a synthetic data sample with respect to the training set was lower than the 2% quantile of the empirical distribution of DCRs of the test set.

Method	Adult	Beijing	Default	Diabetes	Magic	News	Shoppers	Average
CTGAN	0.01 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.03 $\pm$ 0.01	0.0 $\pm$ 0.000	0.09 $\pm$ 0.02	0.0 $\pm$ 0.000	0.0186
TVAE	6.41 $\pm$ 0.11	0.0 $\pm$ 0.000	0.06 $\pm$ 0.01	19.55 $\pm$ 0.17	0.01 $\pm$ 0.01	0.22 $\pm$ 0.02	0.02 $\pm$ 0.01	3.7529
GReaT	0.0 $\pm$ 0.000	1.49 $\pm$ 0.06	12.54 $\pm$ 0.17	—	0.35 $\pm$ 0.04	—	10.8 $\pm$ 0.35	—
STaSy	1.1 $\pm$ 0.04	0.44 $\pm$ 0.04	0.56 $\pm$ 0.06	—	0.28 $\pm$ 0.04	0.25 $\pm$ 0.03	0.27 $\pm$ 0.05	—
CoDi	0.06 $\pm$ 0.01	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.03 $\pm$ 0.01	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0129
TabSyn	1.72 $\pm$ 0.07	0.51 $\pm$ 0.05	1.12 $\pm$ 0.05	0.65 $\pm$ 0.02	0.07 $\pm$ 0.02	0.81 $\pm$ 0.05	1.76 $\pm$ 0.12	0.9486
TabDiff	1.85 $\pm$ 0.07	0.61 $\pm$ 0.05	1.4 $\pm$ 0.06	1.35 $\pm$ 0.05	0.05 $\pm$ 0.02	0.89 $\pm$ 0.05	2.24 $\pm$ 0.09	1.1986
Fully Factorized	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.01 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0014
Fully Factorized (P)	0.04 $\pm$ 0.01	0.08 $\pm$ 0.01	0.0 $\pm$ 0.000	0.01 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0 $\pm$ 0.000	0.0186
ShallowMixture	2.217 $\pm$ 0.064	0.923 $\pm$ 0.06	0.708 $\pm$ 0.055	0.089 $\pm$ 0.009	0.505 $\pm$ 0.068	0.07 $\pm$ 0.01	1.311 $\pm$ 0.1	0.832
TabPC	1.84 $\pm$ 0.09	0.51 $\pm$ 0.04	0.92 $\pm$ 0.06	1.17 $\pm$ 0.05	0.14 $\pm$ 0.03	0.68 $\pm$ 0.04	2.52 $\pm$ 0.15	1.1114

Table 4: Dataset statistics. # Num = number of numerical columns, # Cat = number of categorical columns, # Max Cat = max categories in any categorical column. # Train and # Test refer to the number of samples in the training and test splits respectively.

Dataset	# Rows	# Num	# Cat	# Max Cat	# Train	# Test	Task
Adult	48,842	6	9	42	32,561	16,281	Classification
Default	30,000	14	11	11	27,000	3,000	Classification
Shoppers	12,330	10	8	20	11,097	1,233	Classification
Magic	19,019	10	1	2	17,117	1,902	Classification
Beijing	43,824	7	5	31	39,441	4,383	Regression
News	39,644	46	2	7	35,679	3,965	Regression
Diabetes	101,766	9	27	716	81,412	20,354	Classification